

Question ID: 6aefc52b

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                                        |
|------------|------|---------------|-------------------------------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard <div><div></div><div></div><div></div></div> |

Question

$y = -2.5$

$y = x^2 + 8x + k$

In the given system of equations,  $k$  is a positive integer constant. The system has no real solutions. What is the least possible value of  $k$ ?

Correct Answer: 14

Rationale

The correct answer is 14. It's given by the first equation of the system of equations that  $y = -2.5$ . Substituting  $-2.5$  for  $y$  in the second given equation,  $y = x^2 + 8x + k$ , yields  $-2.5 = x^2 + 8x + k$ . Adding 2.5 to both sides of this equation yields  $0 = x^2 + 8x + k + 2.5$ . A quadratic equation of the form  $0 = ax^2 + bx + c$ , where  $a$ ,  $b$ , and  $c$  are constants, has no real solutions if and only if its discriminant,  $b^2 - 4ac$ , is negative. In the equation  $0 = x^2 + 8x + k + 2.5$ , where  $k$  is a positive integer constant,  $a = 1$ ,  $b = 8$ , and  $c = k + 2.5$ . Substituting 1 for  $a$ , 8 for  $b$ , and  $k + 2.5$  for  $c$  in  $b^2 - 4ac$  yields  $8^2 - 4(1)(k + 2.5)$ , or  $64 - 4(k + 2.5)$ . Since this value must be negative,  $64 - 4(k + 2.5) < 0$ . Adding  $4(k + 2.5)$  to both sides of this inequality yields  $64 < 4(k + 2.5)$ . Dividing both sides of this inequality by 4 yields  $16 < k + 2.5$ . Subtracting 2.5 from both sides of this inequality yields  $13.5 < k$ . Since  $k$  is a positive integer constant, the least possible value of  $k$  is 14.